MODERN ACADEMY FOR ENGINEERING & TECHNOLOGY

Electronic Eng. and Communication Tech. Branch

# DSP LAB Report – Fall 27 – Law 20

**Prepared by:**

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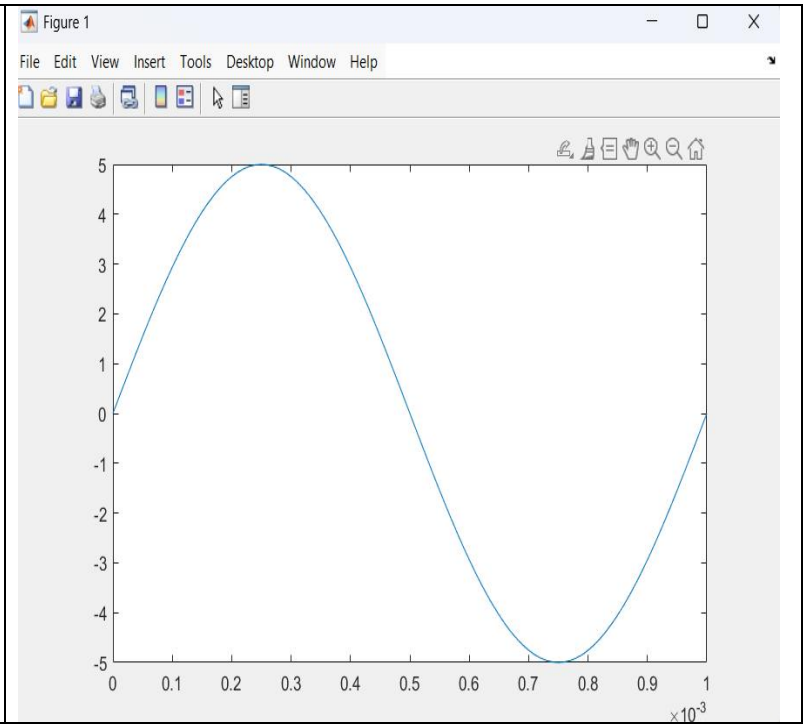
**Under supervision of:**

## Dr. / Samir M. Kamal

**Cairo 2027**

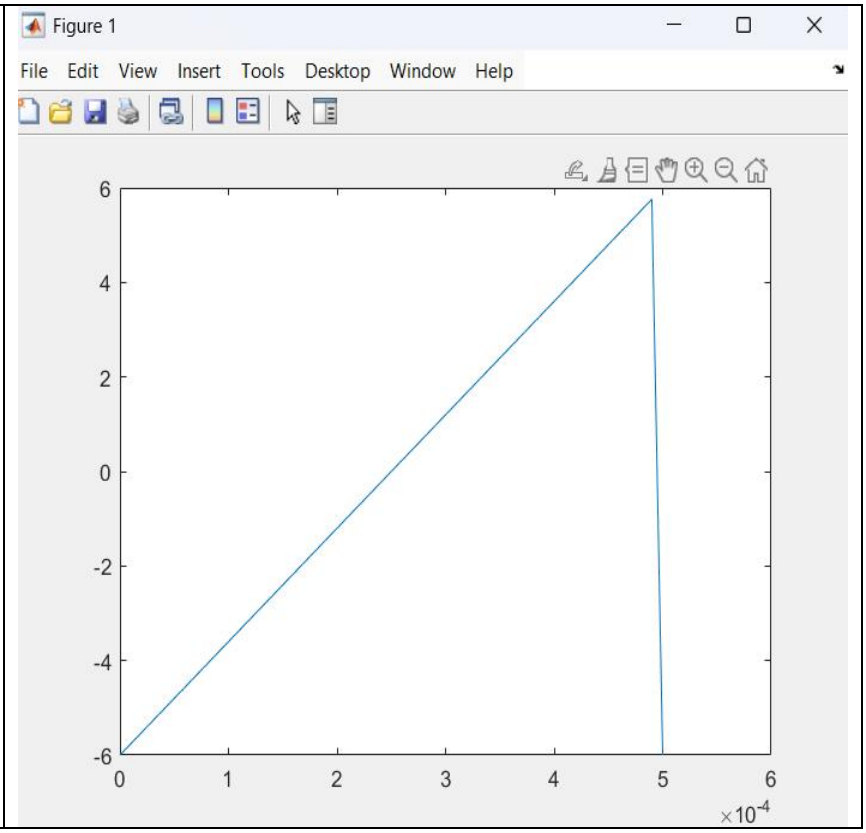
## I. Sine wave with frequency = 1 KHz and amplitude = 5 V

```
fs=100000;
ts = 1/fs;
f = 1000;
t = 1/f;
nts = 0:ts:t;
signal = 5 * sin(2*pi*f*nts);
plot(nts,signal)
```



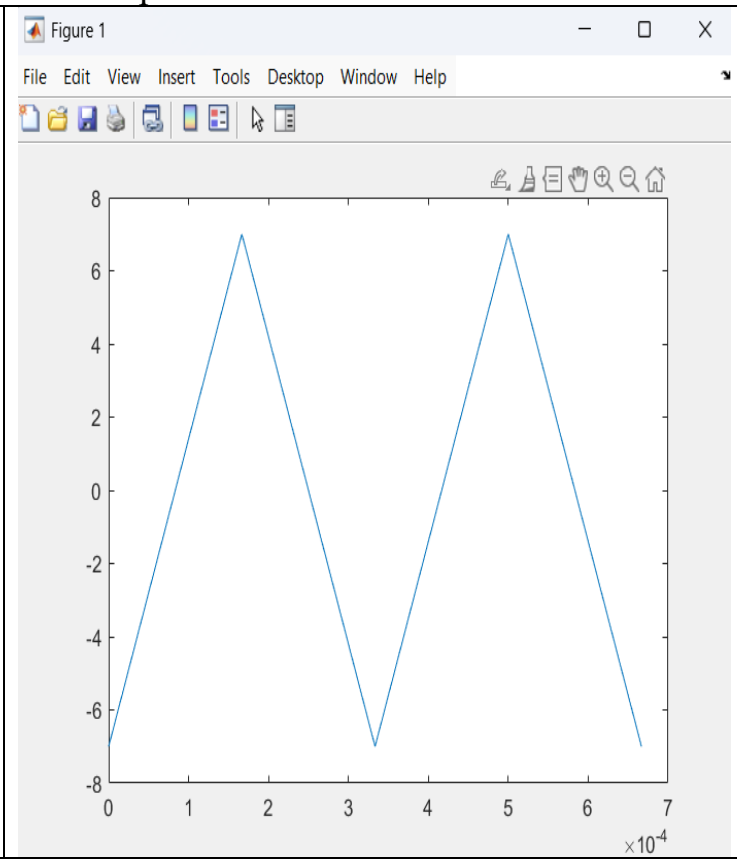

## II. Sawtooth wave with frequency = 2 KHz and amplitude = 6 V

```
fs = 100000;
ts = 1/fs;
f = 2000;
t = 1/f;
nts = 0:ts:t;
signal = 6 *
sawtooth(2*pi*f*nts);
plot(nts,signal)
```



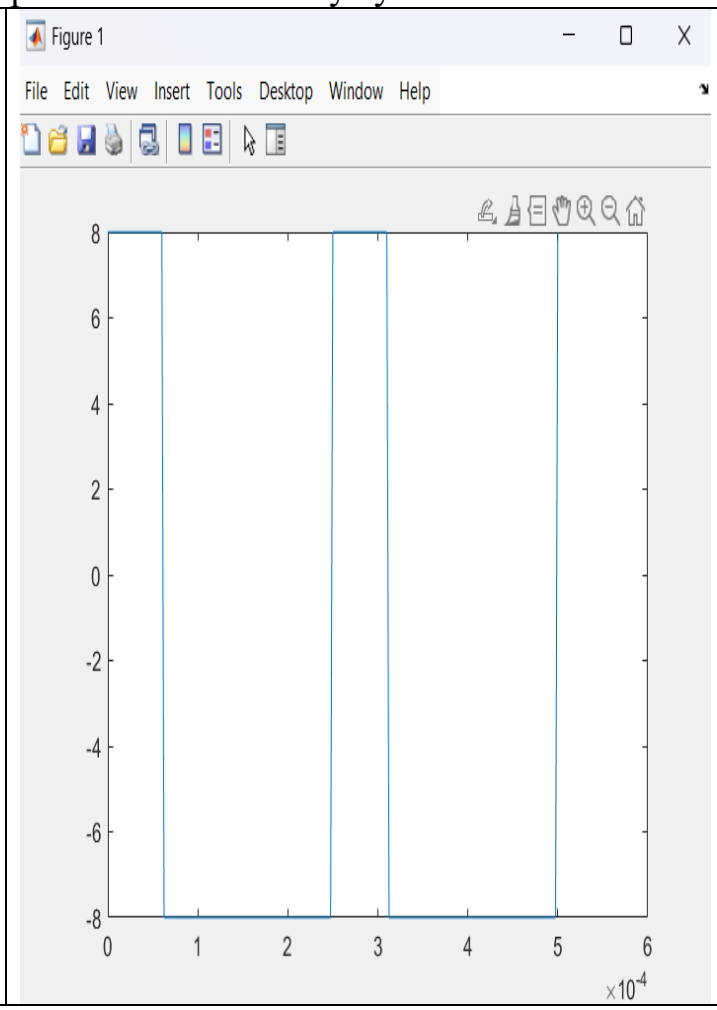
III. Triangle wave with frequency = 3 KHz and amplitude = 7 V

```
fs = 300000;
ts = 1/fs;
f = 3000;
t = 1/f;
nts = 0:ts:2*t;
signal = 7 * sawtooth(2*pi*f*nts, 0.5);
plot(nts,signal)
```




IV. Square wave with frequency = 4 KHz, amplitude = 8 V and duty cycle = 25%

```
fs=400000;
ts = 1/fs;
f = 4000;
t = 1/f;
nts = 0:ts:2*t;
signal=8*square(2*pi*f*nts,25);
plot(nts,signal);
```

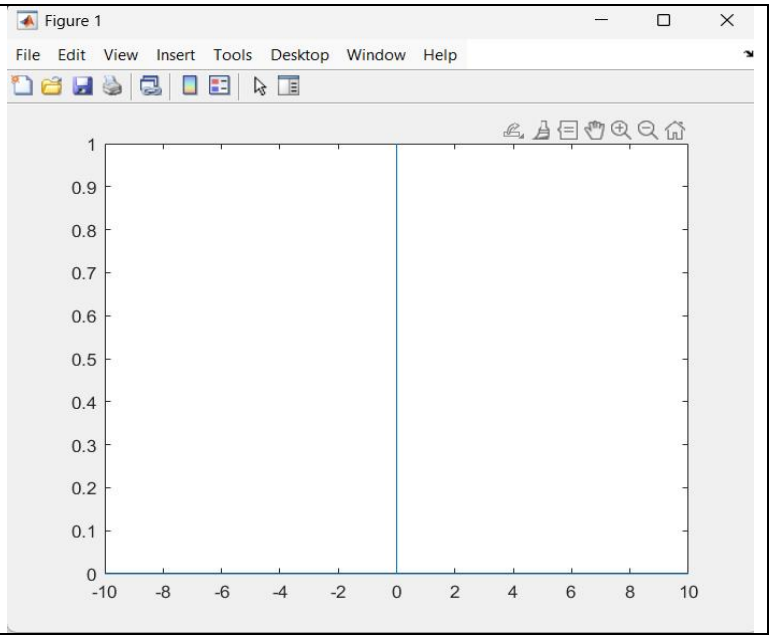


## V. Impulse function time delayed by 1 ms

```
nts = -10:0.001:10;

signal = nts == 0;

plot (nts, signal) ;
```

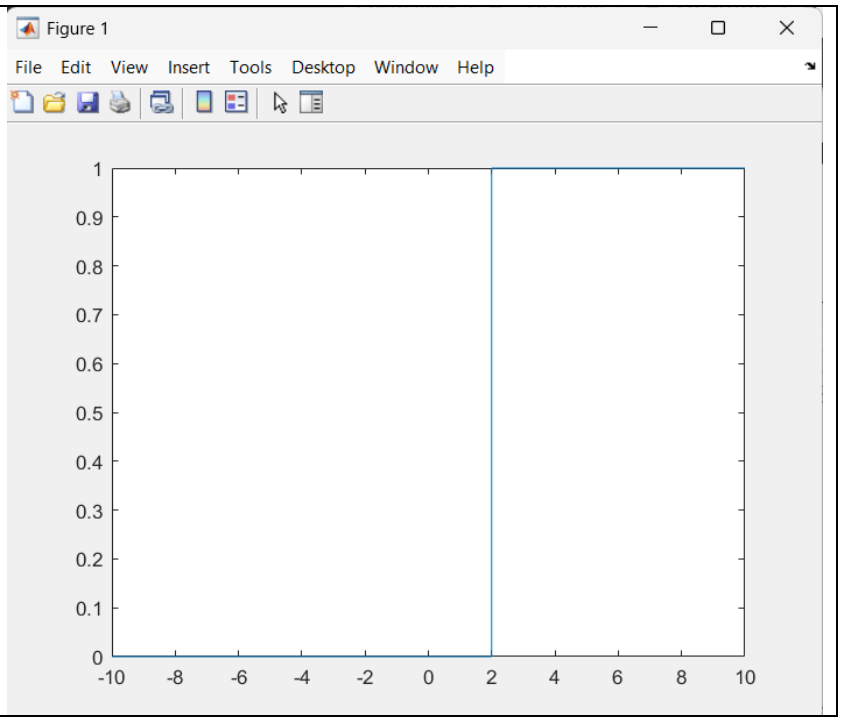



## VI. Unit step function time delayed by 2 ms

```
nts = -10:0.001:10;

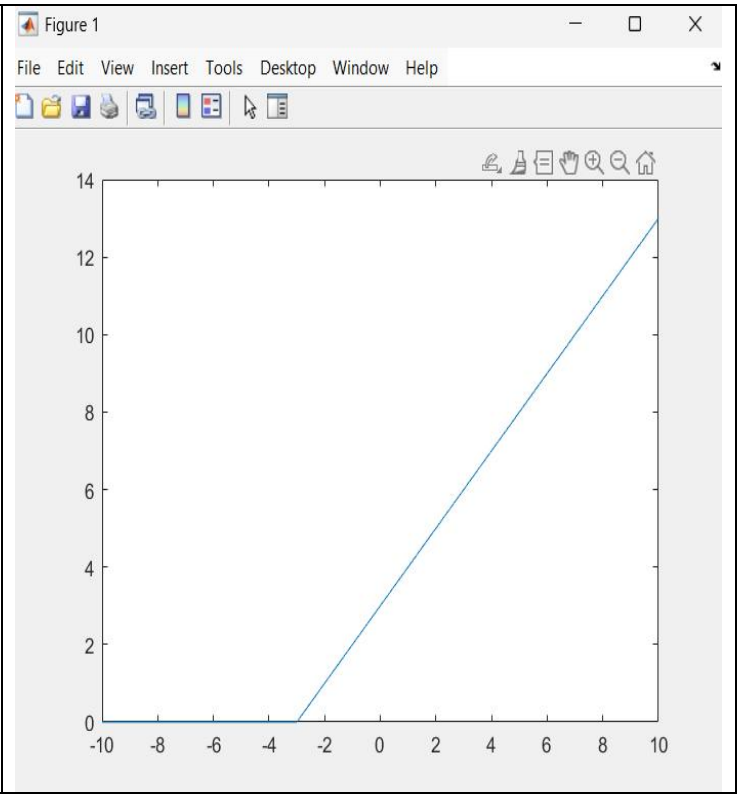
signal = nts-2 >= 0;

plot (nts,signal);
```



VII. Ramp function time advanced by 3 ms

```
nts = -10:0.001:10;
signal = (nts+3).*(nts>=-3) ;
plot (nts,signal)
```

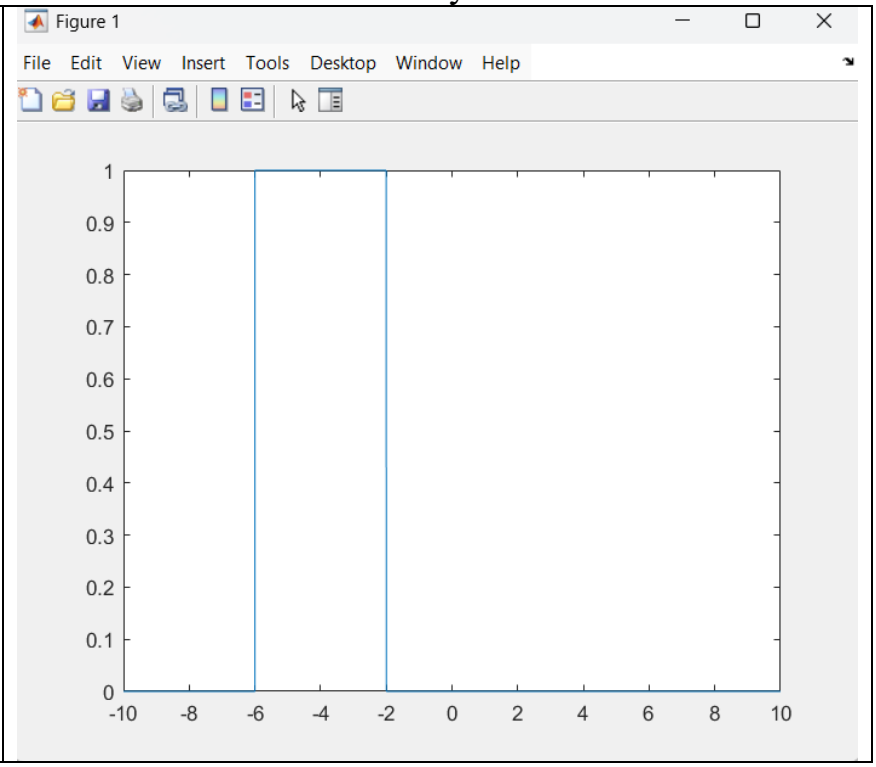



VIII. Rectangular pulse with width = 4 ms and time advanced by 4 ms

```
nts = -10:0.001:10;

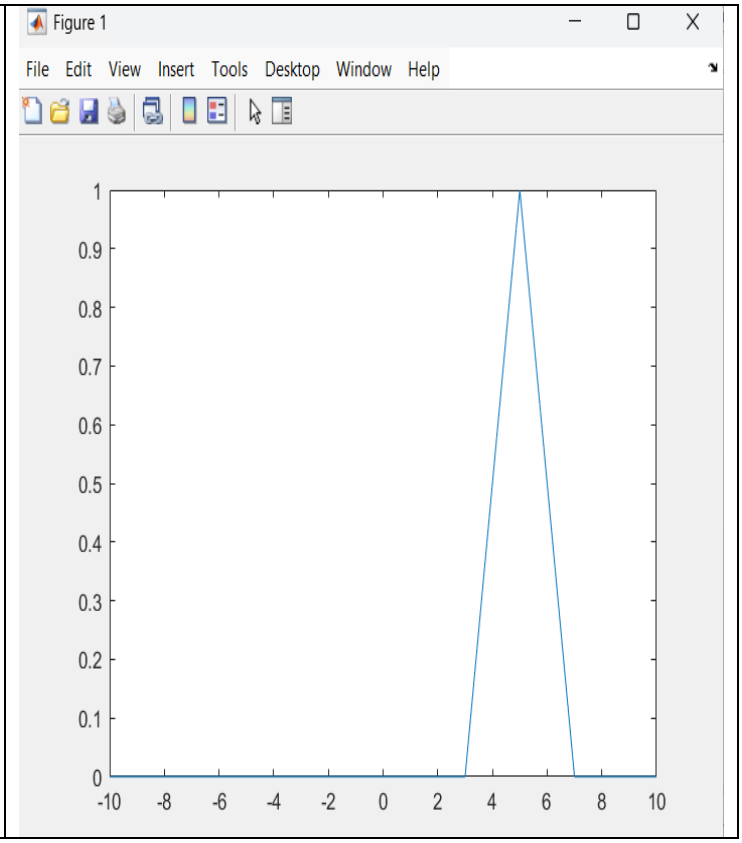
signal = rectpuls (nts + 4, 4);

plot(nts,signal)
```



IX. Triangle pulse with width = 4 ms and time delayed by 5 ms

```
nts = -10:0.001:10;
signal = tripuls (nts - 5, 4);
plot(nts, signal)
```



X. Sinc function where T = 0.5 ms

```
nts = -10:0.001:10;
signal = sinc (nts / 0.5);
plot(nts, signal)
```

